BTS –V–11.14–0928 Reg. No. | | | | | | | | |

C

B. Tech. Degree V Semester Examination November 2014

ME 1506 POWER PLANT ENGINEERING

(2012 Scheme)

Time: 3 Hours Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 x 5 = 40)

I. (a) Define depreciation of power plants and explain any method for calculating annual depreciation.
(b) Write notes on power house equipments.
(c) Illustrate the advantages of hydroelectric power plants.
(d) List out the advantages of gas turbine over steam turbine.
(e) Derive an expression to find chimney diameter at any cross section of the chimney.
(f) List the safety measures for nuclear power plants.
(g) Explain 'solar thermal collection'.
(h) Describe H_2O_2 fuel cells, with neat sketch.

PART B

(4 x 15 = 60)

II. (a) A power station is to supply three region of load whose peak loads are 20MW, 15MW and 25 MW. The annual load factor is 50% and the diversity factor of the load at station is 1.5. Determine the following: (7)
(i) Maximum demand on the station
(ii) Installed capacity suggesting number of unit
(iii) Annual energy supplied.

(b) A base load power station and standby power station share a common load as follows: (8)

Base load station annual output	=	150×10^{6}kWh
Base load station capacity	=	35MW
Maximum demand on base load station	=	30MW
Standby station capacity	=	18MW
Standby station annual output	=	14×10^{6}kWh
Maximum demand (peak load) on standby station	=	15MW

Determine the following parameters for both power stations.
(i) Load factor (ii) Capacity factor (Plant factor)

OR

III. (a) Describe four types of tariffs for electricity. (8)
(b) With a neat sketch, make a note on load curve. Describe different types of load curves. (7)

(P.T.O.)

IV.		Explain different methods for improving thermal efficiency of open cycle gas turbine power plant, with neat sketches.	(15)
		OR	
V.	(a)	Explain the layout of diesel engine power plant, with a neat sketch.	(7)
	(b)	Explain the functions of following systems in a diesel engine power plant	
		(i) Fuel system	(4)
		(ii) Cooling system	(2)
		(iii) Lubricating system	(2)
VI.	(a)	Describe the fluidized bed combustion system, with neat sketch.	(7)
	(b)	Explain dust and ash handling system, with neat sketch.	(8)
		OR	
VII.	(a)	A steam power plant uses coal 5000kg/hr. The heat conversion efficiency is 30% and calorific value of coal is 7000 k.cal/kg. Calculate the electric energy produced per day. (1kWh=869 k.cal).	(7)
	(b)	Explain BWR and PWR in a nuclear power plant, with neat sketches.	(8)
VIII.	(a)	Distinguish between thermoelectric conversion and thermionic conversion system.	(8)
	(b)	Describe Ocean Thermal Energy Conversion (OTEC).	(7)
		OR	
IX.	(a)	Explain MHD power generation, with neat sketches.	(7)
	(b)	Explain different types of solar collectors, with neat sketches.	(8)

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